

Stochastic Processes & Martingales

A compact mathematical research note on random walks, filtrations, martingales, stopping times, quadratic variation, and Brownian motion.

1. Random Walks and Information

Let (X_i) be independent random variables with $P(X_i = 1) = P(X_i = -1) = 1/2$. The symmetric random walk is $S_0 = 0$ and $S_n = X_1 + \dots + X_n$.

The natural filtration $F_n = \sigma(X_1, \dots, X_n)$ represents exactly the information revealed by the first n increments. This distinction between the path and available information is the starting point for dynamic probability.

$$\begin{aligned} S_0 &= 0 \\ S_n &= X_1 + \dots + X_n \\ F_n &= \sigma(X_1, \dots, X_n) \end{aligned}$$

2. Martingales

A process (M_n) is a martingale with respect to (F_n) if it is adapted, integrable, and $E[M_n | F_m] = M_m$ for $m \leq n$.

For the symmetric random walk, future increments are independent of current information and have mean zero. Therefore $E[S_n | F_m] = S_m$, so the random walk is a martingale.

A useful second example is the exponential martingale $M_n(\theta) = \exp(\theta S_n) / (\cosh(\theta))^n$. The denominator removes the expected exponential growth.

$$E[S_n | F_m] = E[S_m + X_{m+1} + \dots + X_n | F_m] = S_m$$

3. Stopping Times

A random time τ is a stopping time if the event $\{\tau \leq n\}$ is known at time n for every n . A first hitting time is a stopping time; the time of the future maximum is generally not.

This is the mathematical version of avoiding look-ahead bias: a valid rule must be measurable with respect to the information available when the decision is made.

$$\tau = \inf\{n \geq 0 : |S_n| \geq a\}$$

4. Optional Stopping

If (M_n) is a martingale and τ is a bounded stopping time, then $E[M_\tau] = E[M_0]$. For a symmetric random walk starting at zero, this gives $E[S_\tau] = 0$.

The message is precise: under suitable conditions, stopping a fair game does not create positive expectation. The conditions are essential; without boundedness or integrability, the conclusion can fail.

5. Brownian Motion

Brownian motion W_t starts at zero, has independent increments, satisfies $W_t - W_s \sim \text{Normal}(0, t-s)$, and has continuous sample paths.

It is the canonical continuous-time limit object for accumulated independent shocks.

6. Quadratic Variation

For a partition $0 = t_0 < \dots < t_n = T$, quadratic variation is the sum of squared increments. For Brownian motion, this converges to T .

This is mathematically striking: Brownian paths are continuous but not smooth. Their squared increments do not vanish; they accumulate into elapsed time.

$$\sum_k (W_{t_{k+1}} - W_{t_k})^2 \rightarrow T$$

7. From Random Walks to Brownian Motion

Define $W_n(t) = S_{\text{floor}(nt)} / \sqrt{n}$. At $t = 1$, the central limit theorem gives convergence toward a standard normal variable.

Donsker's invariance principle extends this endpoint convergence to process-level convergence: the scaled and interpolated random walk converges in distribution toward Brownian motion.

$$W_n(t) = S_{\text{floor}(nt)} / \sqrt{n}, \quad 0 \leq t \leq 1$$

8. Why This Matters

This note is not an investment memo. Filtrations explain information sets. Martingales explain conditional fairness. Stopping times explain legitimate timing rules. Quadratic variation explains pathwise roughness. Scaling limits explain how discrete stochastic systems can converge to continuous models.

The correct reading is: this is the mathematical foundation underneath systematic research.